

# Multi-level contour combination features for shape recognition

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## ABSTRACT

We present a novel multi-level contour combination feature for shape recognition. This combination feature effectively solves large intra-class changes and nonlinear deformations of object shapes, thereby enhancing the performance of shape recognition. First, we divide the shape contour into two levels: the sampling points and the contour fragments, where sampling points are used to describe the detailed information of a shape and contour fragments are used to represent the global feature of a shape. Second, we employ the Fisher vector (FV) approach to encode the local sampling point feature and contour fragment feature as high-level characteristics. Finally, we combine the high-level characteristics after FV encoding and perform shape recognition through a linear support vector machine (SVM) model. The proposed method has been assessed on three benchmark shape datasets, including the Animal, MPEG-7, and ETH-80 datasets. Our method achieves 92.70%, 99.26% and 98.32% classification accuracy on the Animal, MPEG-7, and ETH-80 datasets, respectively. In addition, our method can also be applied to the classification of objects in real-world scenes. We combine the Weizmann Horse and the ETHZ Cow real-world scene datasets, and our method achieves 99.25% classification accuracy on the combined dataset. The recognition results of our approach are better than prior state-of-the-art shape recognition methods, which demonstrate the effectiveness and superiority of our approach.

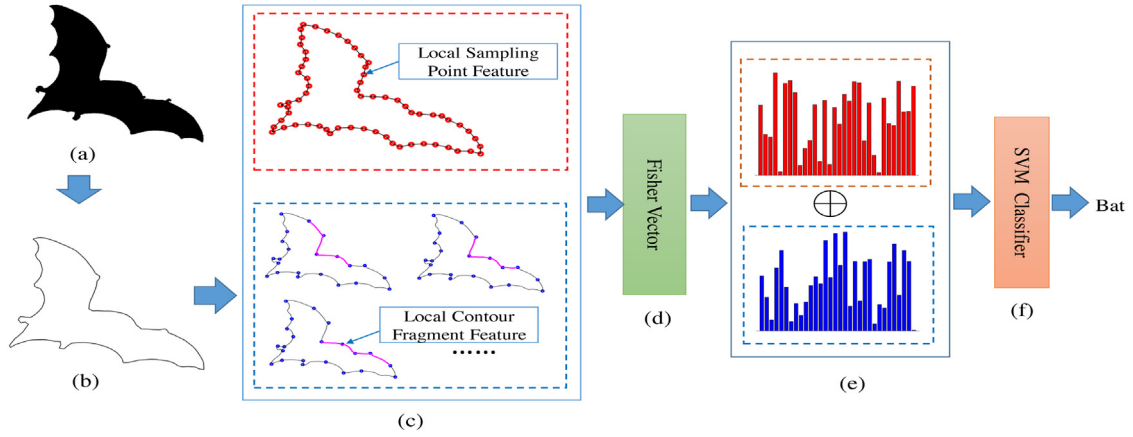
## 1. Introduction

Shape is a very important visual attribute of an object in pattern recognition and image analysis, which provides very important information for the classification and recognition of the target object. Compared with the appearance characteristics of the object such as texture and color, shape is a very stable feature and is hardly affected by changes in object color, texture or illumination. Therefore, shape recognition has always been a hot research direction in the field of pattern recognition and computer vision, and it has a wide range of applications in object detection, image retrieval and object recognition. Shape recognition is customarily regarded as a classification issue. For a given test shape, we predict its category label using a group of training shapes and their class labels. The main difficulty of shape recognition comes from the large intra-class changes and nonlinear deformations (such as articulation changes and noise) of objects. Therefore, the research of shape recognition has mainly focused on how to build an informative and distinctive shape descriptor in the past decade (Ling, 2007; Xu et al., 2009; Wang and Gao, 2014; Shen et al., 2018; Yang and Yu, 2019; Wang and Gao, 2020; Yang and Yu, 2021; Yang, 2021; Yang et al., 2022).

A large number of shape descriptors have been proposed, which can be separated into boundary-based and region-based descriptors (Mingqiang et al., 2008; Kurnianggoro et al., 2018). The boundary-based descriptors only employ the boundary information of an object to extract shape feature, such as the shape context (SC) (Belongie et al., 2002), contour flexibility (Xu et al., 2009), binary angular pattern (Hu et al., 2014), multiscale triangular centroid distance (MTCD) (Yang et al., 2016) and multiscale Fourier descriptor (MFD) (Yang and Yu, 2019). The region-based descriptors employ the whole region of an object to extract shape feature, such as Zernike moment (Khotanzad and Hong, 1990), Radon composite feature (Chen and Chen, 2008), structure integral transform (Wang and Gao, 2016) and contour covariance (Yu et al., 2019). However, these shape descriptors are low-level shape description approaches, and they are still difficult to effectively solve the large intra-class changes and nonlinear deformations of object shapes, resulting in low accuracy of shape recognition. At the same time, they use one-to-one matching methods in shape recognition, such as dynamic programming method, so as to get the dissimilarity between each two shapes, and then use the nearest neighbor (NN) model for recognition. The principal shortcoming of these approaches is that it requires a very high matching cost to perform shape matching one by one for larger training samples.

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**Fig. 1.** The pipeline of our shape recognition approach based on multi-level contour combination features: (a) original shape; (b) The contour (black curve) of (a); (c) the multi-level contour features of the shape (a), which includes the sampling points and contour fragments two levels. Here, we use the triangular feature to describe the sampling point and adopt the curvature, tangential angle and Fourier descriptor to describe the contour fragment; (d) the Fisher vector is used to encode the local features; (e) the shape features obtained after encoding the sampling points and contour segments, respectively, where the sign  $\oplus$  represents the combination of two features; and (f) the linear SVM classifier is used for shape recognition based on the combined shape feature. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

To effectively solve the problems of low-level shape descriptors and enhance the performance of shape recognition, we propose a novel multi-level contour combination feature for shape recognition. First, we divide the shape contour into two levels: sampling points and contour fragments. The features of sampling points are used to capture the detailed characteristics of a shape, while the features of contour fragments are used to capture the global information of a shape. The shape information of an object can be more effectively described by combining the above two-level features, and the better shape distinguishing ability can be obtained. Second, we employ the Fisher vector method (Sánchez et al., 2013) to encode the local features of sampling points and contour fragments into high-level shape feature, respectively. Finally, we combine the encoded high-level shape information and perform shape recognition through a linear SVM classifier.

The pipeline of the proposed shape recognition method based on multi-level contour combination features is presented in Fig. 1. First, we extracted the contour of an object for a binary image, which is presented in Fig. 1(b). Next, we divided the shape contour into two levels: sampling points and contour fragments and used triangular feature to describe the sample points and contour fragments, as shown in Fig. 1(c). We directly sample the whole shape contour uniformly, which is shown by the red dots in Fig. 1(c). To acquire the contour fragments of an object, we first used the discrete curve evolution (DCE) method (Latecki and Lakämper, 1999) to obtain some critical points, and then extracted the contour fragments between different critical points. The blue points in Fig. 1(c) represent the critical points obtained by the DCE method, and the magenta color in Fig. 1(c) represents the extracted contour fragment. Here, we extracted the curve of five critical points as the contour fragment. Third, we used the Fisher vector method to separately encode the local sampling point features and the local contour fragment features to obtain high-level characteristics, which is presented in Fig. 1(d). Fourth, we combined the features after Fisher vector encoding as the final shape representation, which is presented in Fig. 1(e). This shape feature can effectively describe the local and global information of an object, and can remain robust to large intra-class changes and nonlinear deformations of objects. Finally, we adopt the linear SVM model based on the combined shape feature for shape recognition.

The remainder of the work is organized as follows. In the second section, we briefly review the related work. In Section 3, we present the proposed multi-level contour combination description approach in detail. Section 4 presents our experimental results on some benchmark shape datasets. Finally, we conclude this article.

## 2. Related work

A large number of shape recognition methods have been proposed in the past few decades (Sun and Super, 2005; Ling, 2007; Daliri and Torre, 2008; Bai et al., 2009; Wang et al., 2010, 2012, 2014; Ke and Li, 2014; Ramesh et al., 2015; Bicego and Lovato, 2016; Li et al., 2016; Govindaraj and Sudhakar, 2018; Shen et al., 2018; Patel et al., 2019; Yang et al., 2020; Mirehi et al., 2021), which can be separated into exemplar-based and model-based approaches. Next, we briefly review some important shape recognition methods.

The exemplar-based approaches (Sebastian et al., 2004; Ling, 2007; Daliri and Torre, 2008; Macrini et al., 2011; Wang et al., 2012; Bicego and Lovato, 2016; Govindaraj and Sudhakar, 2018) first extract an effective shape feature, then use the shape matching algorithm to calculate the dissimilarity between each two shapes, and finally adopt the NN model for shape recognition. Belongie et al. (Belongie et al., 2002) proposed a SC descriptor, which describes the spatial distribution relationship between a certain point on the contour and all other points on the contour. This descriptor contains rich information and has a very strong ability to describe shapes. A well-known extension of the SC descriptor is the inner-distance shape context (IDSC) (Ling, 2007), which employs the inner-distance in place of Euclidean distance in SC to effectively tackle the articulation variation of object shapes. Daliri et al. (Daliri and Torre, 2008) combined the SC descriptor with string of symbols to improve the performance of shape recognition. They first converted the shape contour into a symbol string representation, then used the edit distance to measure the dissimilarity between the symbol strings, and finally realized the shape recognition based on the dissimilarity. Wang et al. (Wang et al., 2012) presented a height function description method for shape recognition. They adopt the dynamic programming method to compute the dissimilarity between each two shapes, and then use the calculated dissimilarity to identify the shape. To solve the large deformation of an object, some researchers consider using the skeleton of an object for shape recognition. Sebastian et al. (Sebastian et al., 2004) presented a shock graph description approach to identify shapes, which is very effective for large deformations of objects. Macrini et al. (Macrini et al., 2011) proposed a shape recognition approach using the bone graph and acquired high recognition accuracy. However, the skeleton-based shape representation method is not as effective as the contour-based method for the recognition of general object's shapes. To further enhance the accuracy of shape classification, some recent shape description approaches have been presented. Bicego et al. (Bicego and Lovato, 2016) proposed a bioinformatics method to recognize 2D shapes. First, they converted

shape matching into a sequence matching problem. Then, they employed the sequence matching method in bioinformatics to measure the dissimilarity between two shapes. Finally, they used the NN model to complete the shape recognition task. Govindaraj et al. (Govindaraj and Sudhakar, 2018) proposed a hexagonal grid-based triangulated tessellation characteristic for shape recognition. This descriptor achieves better shape classification performance, and at the same time it has lower time complexity. Nevertheless, the exemplar-based approaches mainly consider how to construct an effective low-level shape descriptor, which is not robust enough for large intra-class changes and nonlinear deformations of object shapes. At the same time, the shape matching of these methods is very time-consuming while the amount of training data is very large.

The model-based method (Sun and Super, 2005; Bai et al., 2009; Wang et al., 2010; Daliri and Torre, 2010; Wang et al., 2014; Ramesh et al., 2015; Shen et al., 2018; Yang et al., 2020) learns an effective classification model for shape recognition, instead of recognizing shapes based on shape matching algorithm in exemplar-based methods. Sun et al. (Sun and Super, 2005) presented a shape recognition framework using the Bayesian classifiers, which employs normalized contour segments as input features to recognize the shape of objects. Bai et al. (Bai et al., 2009) proposed a Gaussian mixture model (GMM) approach for shape recognition. This approach effectively combines the contour features and skeleton information of an object to handle the articulation changes of object shapes, and it obtains good recognition results. Wang et al. (Wang et al., 2010) presented a shape tree descriptor using the junction nodes for shape recognition. This method can capture the structural information of the shape well, and can identify objects with large deformation. Daliri et al. (Daliri and Torre, 2010) proposed a shape classification method using the kernel-edit distance. First, they transformed the shape boundary into a symbolic string representation. Then, they employed the normalized edit distance to measure the dissimilarity between symbol strings. Finally, they transformed the edit distance into an appropriate kernel function, and used the SVM model for shape classification. Wang et al. (Wang et al., 2014) introduced a shape descriptor named bag of contour fragments (BCF) for shape recognition. This approach adopted the bag of words (BoW) model to construct an effective and robust shape descriptor, and obtained good classification results on shape recognition task. Ramesh et al. (Ramesh et al., 2015) presented a shape classification framework using the BoW model, which can combine invariant features and context information to learn an efficient and robust shape descriptor. This method has achieved good results on the task of shape classification. Shen et al. (Shen et al., 2018) proposed a feature pooling method to adaptively adjust the combined weight of input features, which can effectively combine the advantages of different shape descriptors. This method combines the characteristics of the BCF and bag of skeleton-associated contour parts (BoSCP) for shape recognition, and obtains a high recognition accuracy.

The most relevant to this article is the work proposed by Yang et al. (Yang et al., 2020). They proposed a shape classification method based on contour-based mid-level representation (CMLR). This method builds an effective mid-level contour feature and improves the accuracy of shape classification. However, we have three differences in comparisons with the work of Yang et al. (Yang et al., 2020). First, the proposed method in this paper not only considers the sampling points of a shape, but also considers the contour segment characteristics of a shape. The sampling point feature can effectively characterize the detailed information of a shape, and the contour segment can characterize the global feature of a shape. It can significantly improve the distinguishing ability of shape by fusing the features of sampling points and contour segments. Second, we only use the triangle area  $TAR$  and the two side lengths and intersection angle of a triangle  $TASL$  in the feature representation of sampling points. We did not use the triangle center distance feature  $TCD$ , which reduces the dimension of the feature and improves the performance of shape recognition. Finally,

the proposed method in this paper improves the recognition accuracy of shape classification on several standard shape datasets. Therefore, the work of this paper is based on the extension of the contour-based mid-level shape representation (Yang et al., 2020), which improves the recognition performance of shape classification by effectively fusing the sampling points and contour segment information of a shape.

In recent years, deep learning has become the most important technical method in computer vision (Guo et al., 2016; Porikli et al., 2018), speech recognition (Tan et al., 2018) and natural language processing (Young et al., 2018). Some researchers have considered applying the deep learning techniques for shape analysis. To our knowledge, the earliest application of deep learning technology to the field of shape analysis is the shape Boltzmann machine method proposed by Eslami et al. (Eslami et al., 2014). They use this method to generate shape samples, which can generate realistic shape samples different from the original binary training shapes. Therefore, this method has a very wide application prospect. Ke et al. (Ke and Li, 2014) presented a convolutional neural network model to learn robust shape descriptor, and then adopted this model to cope with the rotation changes of object shapes. Li et al. (Li et al., 2016) presented a neural network optimization approach, which adopts the stochastic gradient Markov Chain Monte Carlo (SG-MCMC) to learn the weight of neural network, so as to accurately estimate the uncertainty of the model. At the same time, this method has achieved good classification accuracy in shape recognition tasks. Patel et al. (Patel et al., 2019) presented a data augmentation approach using the part analysis. First, they used the data augmentation approach to generate a large number of shape data, and then used the generated shape data to train the convolutional neural network (CNN) model for shape recognition. This method obtains high recognition accuracy in shape classification task. Nevertheless, the deep learning approach has some shortcomings, such as requiring a large number of computing resources, massive training data, and insufficient interpretability of the learned model. Recently, some researchers have made useful attempts to address these shortcomings (Shin et al., 2016; Zhuang et al., 2020; Frankle and Carbin, 2019; Lin et al., 2022; Shi et al., 2020; Zhou et al., 2021). For example, transfer learning methods have been proposed for model learning to avoid the problem of requiring massive training data. For a comprehensive overview of transfer learning, please refer to literature (Shin et al., 2016; Zhuang et al., 2020). The transfer learning method generally adopts model fine-tuning, but its recognition accuracy is lower than that of the model trained from scratch. Meanwhile, to solve the problem of requiring a large number of computing resources, some researchers have used knowledge distillation and pruning strategies (Frankle and Carbin, 2019; Lin et al., 2022) to reduce the number of parameters of deep models. In addition, recent research found that CNN model has the problem of texture bias, and Shi et al. (Shi et al., 2020) proposed an Informative Dropout method to solve this problem. This method detects texture and shape by the local self-information in an image and uses a dropout-like algorithm to decorelate the model output from the local texture, which improves the interpretability of deep CNN models and solves the problem of texture bias. Moreover, this method improves the robustness of the model under various scenarios, including domain generalization, few-shot classification, and adversarial perturbation through extensive experiments. To further improve the generalization ability of deep models, Zhou et al. (Zhou et al., 2021) proposed a Mixstyle method. The method significantly improves the generalization performance of the deep model by using style information, and demonstrates strong potential for application in category classification, instance retrieval, and reinforcement learning tasks. However, the recognition accuracy of these methods is still relatively low when applied to 2D shape recognition tasks. Compared to the deep learning approach, our method only needs a small amount of data and less computing resources to learn, and the resulting model has strong interpretability. Subsequent experimental results prove that our approach obtains the highest classification accuracy in shape recognition, and the recognition performance of our method exceeds traditional shape recognition approaches as well as deep learning-based shape recognition methods.

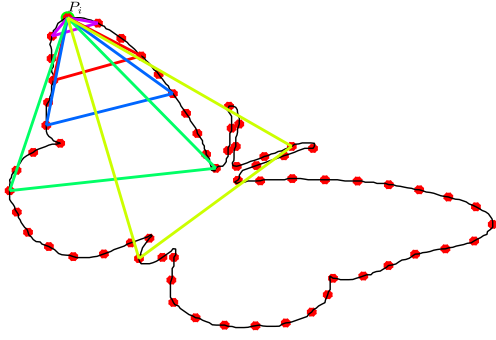


Fig. 2. The triangular descriptor of point  $P_i$  for the contour  $S$ . We obtain  $N = 64$  points for the  $S$  and the  $P_i$  is described by five triangles ( $T_s = 5$ ).

### 3. Background (triangular feature and Fisher vector)

In this section, we briefly review the background of triangular features and Fisher vector. This is because they need to be used in our proposed shape recognition framework. The proposed multi-level contour combination features consider two levels: sampling points and contour segments. The contour sampling point is characterized by triangular features, which is introduced in Yang et al. (2020). Then, we use Fisher vector method (Perronnin et al., 2010; Sánchez et al., 2013) to encode the features of local sampling points and local contour segments to learn a high-level robust feature representation. Fisher vector is a very popular encoding method that is often used for feature accumulation.

#### 3.1. Triangular feature

We assume that the extracted shape contour  $S$  consists of  $N$  boundary points  $P_n = \{(x_n, y_n), n = 1, 2, \dots, N\}$ . For each sampling point  $P_i$  of the  $S$ ,  $T_s$  triangles are used to describe it.  $T_s$  denotes the number of triangles at different scales, and it takes  $\lfloor \log_2(N/2) \rfloor$ . More specifically, it can be shown in Fig. 2.

In what follows, each point  $P_i$  is associated with  $T_s$  triangles  $Tr_i^1, \dots, Tr_i^{T_s}$ , and the triangle  $Tr_i^k$  is defined by the boundary points  $P_{i-l(k)}$ ,  $P_i$  and  $P_{i+l(k)}$  ( $i \in [1, N]$ ,  $k \in [1, T_s]$ ).  $l(k)$  denotes the logarithmic distance between points of the  $S$  where  $l(T_s) < N/2$ . Three triangular features associated to each point  $P_i$  are introduced in Yang et al. (2020), which is given by

$$\begin{aligned} TAR(P_i) &= (TAR(Tr_i^1), \dots, TAR(Tr_i^{T_s})), \\ TCD(P_i) &= (TCD(Tr_i^1), \dots, TCD(Tr_i^{T_s})), \\ TASL(P_i) &= (TASL(Tr_i^1), \dots, TASL(Tr_i^{T_s})). \end{aligned} \quad (1)$$

Here,  $TAR(Tr_i^k)$  is the signed area of the  $Tr_i^k$ ,  $TCD(Tr_i^k)$  represents the distance from point  $P_i$  to the centroid of the  $Tr_i^k$ , and  $TASL(Tr_i^k)$  consists of two side lengths and included angle at point  $P_i$  of the  $Tr_i^k$ . Let  $(x_{i-l(k)}, y_{i-l(k)})$ ,  $(x_i, y_i)$ , and  $(x_{i+l(k)}, y_{i+l(k)})$  denote the coordinates of the  $P_{i-l(k)}$ ,  $P_i$  and  $P_{i+l(k)}$ , respectively. First of all, we compute the signed area of the  $TAR(Tr_i^k)$ , as expressed as follows:

$$TAR(Tr_i^k) = \frac{1}{2} \begin{vmatrix} x_{i-l(k)} & y_{i-l(k)} & 1 \\ x_i & y_i & 1 \\ x_{i+l(k)} & y_{i+l(k)} & 1 \end{vmatrix}. \quad (2)$$

The value of  $TAR$  can reflect the uneven characteristics of a shape boundary. Then, we compute the two side lengths and included angle at  $P_i$  of the  $Tr_i^k$ . Next, we calculate the central distance  $TCD(Tr_i^k)$  of  $Tr_i^k$ , which is given by the following expression:

$$TCD(Tr_i^k) = \sqrt{(x_i - x_{g_{ik}})^2 + (y_i - y_{g_{ik}})^2}. \quad (3)$$

Here,  $(x_{g_{ik}}, y_{g_{ik}})$  is the coordinates of the centroid point  $g_{ik}$ , with  $i \in [1, N]$  and  $k \in [1, T_s]$  of triangle  $\triangle P_{i-l(k)} P_i P_{i+l(k)}$ .

Lastly, we calculate the two side lengths and intersection angle at  $P_i$  of  $Tr_i^k$ . Let  $L_i^{1k}$ ,  $L_i^{2k}$  and  $L_i^{3k}$  represent the side lengths of the  $Tr_i^k$ , and sort them to satisfy the relationship  $L_i^{1k} \leq L_i^{2k} \leq L_i^{3k}$ . Let  $M_i^k = \frac{L_i^{1k}}{L_i^{3k}}$  and  $N_i^k = \frac{L_i^{2k}}{L_i^{3k}}$ . The  $\alpha_i^k$  is the included angle at point  $P_i$  of the  $Tr_i^k$ . Hence, the  $TASL(Tr_i^k)$  is defined as follows:

$$TASL(Tr_i^k) = (M_i^k, N_i^k, \alpha_i^k). \quad (4)$$

#### 3.2. Fisher vector method

The Fisher Vector method (Perronnin et al., 2010; Sánchez et al., 2013) is an effective local feature encoding method, which can effectively combine local features to construct robust high-level features. This method is first proposed for the recognition of natural images, which achieves good recognition accuracy in natural image classification. Let  $X = \{x_i, i = 1, 2, \dots, T\}$  be the set of  $T$  local descriptors extracted from an image. Here, we suppose that  $X$  can be generated by the probability density function  $u_\lambda$  with parameter  $\lambda$ . We choose  $u_\lambda$  to be a GMM in the light of literature (Perronnin et al., 2010; Sánchez et al., 2013):  $u_\lambda(x) = \sum_{j=1}^K w_j u_j(x)$ , where  $K$  is the number of ingredients in GMM. The parameter  $\lambda$  is given by  $\lambda = \{w_j, \mu_j, \Sigma_j, j = 1, \dots, K\}$ , where  $w_j$ ,  $\mu_j$  and  $\Sigma_j$  represent the mixture weight, average vector and covariance matrix of Gaussian  $u_j$ , respectively. At the same time, we assume that the  $\Sigma_j$  is a diagonal matrix and  $\sigma_j^2$  is the variance vector. The maximum likelihood (ML) estimation is adopted to acquire the GMM  $u_\lambda$ , which is trained on a large number of images. Let  $\gamma_i(j)$  denotes the soft allocation of each local feature  $x_i$  to Gaussian  $j$ :

$$\gamma_i(j) = \frac{w_j u_j(x_i)}{\sum_{k=1}^K w_k u_k(x_i)}. \quad (5)$$

Suppose that the dimension of  $x_i$  is  $D$ , we can get two gradients of size  $D$  with respect to mean  $\mu_j$  and standard deviation  $\sigma_j$ . Their mathematical expression are given by

$$\mathcal{G}_{\mu,j}^X = \frac{1}{N \sqrt{w_j}} \sum_{i=1}^N \gamma_i(j) \left( \frac{x_i - \mu_j}{\sigma_j} \right), \quad (6)$$

$$\mathcal{G}_{\sigma,j}^X = \frac{1}{N \sqrt{2w_j}} \sum_{i=1}^N \gamma_i(j) \left[ \frac{(x_i - \mu_j)^2}{\sigma_j^2} - 1 \right], \quad (7)$$

where the division between vectors is an term-by-term operation. The two gradient vectors  $\mathcal{G}_{\mu,j}^X$  and  $\mathcal{G}_{\sigma,j}^X$  ( $j = 1, \dots, K$ ) are concatenated to obtain the high-level characteristic  $FS_\lambda^X$ , and the size of this feature is  $2KD$ .

### 4. The proposed multi-level contour description method

We introduce the proposed multi-level contour combination features in this section. First, we present how to divide the shape contour into two levels, including sample points and contour fragments. Second, we introduce the local feature representation methods to describe the sampling points and contour fragments, respectively. Third, we present how to use the Fisher vector encoding approach to obtain high-level shape feature, and combine different levels of high-level characteristics. Finally, we adopt the linear SVM model for shape classification using the multi-level contour combination characteristics.

#### 4.1. Shape contour division

We adopt the edge tracking method to acquire the boundary  $S$  of a binary shape. Let  $S(t) = (x(t), y(t))$  denotes the  $S$  of a shape parameterized by  $t \in [0, 1]$ . We divide the  $S$  of a shape into two levels: sampling points and contour fragments. We directly use the uniform sampling to obtain  $N$  points of the outer boundary  $S = \{P_1, \dots, P_N\}$ .

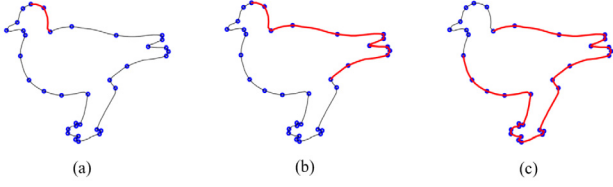


Fig. 3. Three exemplary contour segments (red curve) extracted from a bird image: (a) short-range contour; (b) middle-range contour; (c) long-range contour information in shape. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

Here, each sampling point  $P_i$  ( $i \in [1, N]$ ) represents a local feature, which is used to effectively represent the detailed information of a shape.

To obtain a large number of meaningful contour fragments, we use the DCE method to acquire a simplified polygon on  $S$  expressed as  $U = (u_1, \dots, u_T)$ , and  $T$  is the number of vertices, which is automatically calculated by a threshold parameter  $\tau$ . In the experiment, we set the parameter  $\tau = 0.2$ .  $U$  contains the critical points on  $S$ . The blue point in Fig. 1(c) is the critical points obtained by the DCE method (Latecki and Lakämper, 1999). Given the outer contour  $S$  of a shape, its contour fragment set is represented by  $C(S)$ , which are the curve segments between each pair of critical points ( $u_i, u_j$ ). Let  $c_{ij}$  denote the contour segment between  $u_i$  and  $u_j$ , so we have

$$C(S) = \{c_{ij} = (u_i, u_j), i \neq j, i, j \in [1, T]\}. \quad (8)$$

Note that  $u_i$  and  $u_j$  do not need to be adjacent to each other, so we have  $S = c_{ij} \cup c_{ji}$ . Among them, one represents the contour segment and the other represents its counterpart. Since we extract the contour segments between each pair of critical points,  $C(S)$  has rich information of shape contour  $S$ . All the contour segments extracted from a shape involve the multi-scale information of an object, which can be expressed as short-range, middle-range and long-range information. Fig. 3 gives an example to explain these contour segments. The blue points in the figure represent the critical points extracted by the DCE method (Latecki and Lakämper, 1999), and the red curves represent the extracted contour segments. Fig. 3(a) represents the short-range contour segment, and Fig. 3(b) represents the middle-range contour segment. Fig. 3(c) represents the long-range contour segment.

#### 4.2. Local feature representation

After obtaining the two levels of sampling points and contour segments of the shape contour, we need to use an effective low-level shape descriptor to characterize the sample points and contour segments of an object's shape, respectively. Each sampling point in the shape of an object is considered as a local feature, and the extracted local contour segment is also considered as a local feature. In the following, we introduce the low-level shape descriptor which is used to characterize the sampling points and local contour segments of a shape.

First, we describe the local point feature of a shape. For the representation of local sampling points, we use triangular features to describe them. Unlike the literature (Yang et al., 2020), we use only two triangular features, TAR and TASL, to characterize the contour sampling points. This is because we find that the TCD feature do not lead to performance gains and also increase the dimension of the sampled point features. For any sampling point  $P_i$  in the contour, we use  $TAR(P_i)$  and  $TASL(P_i)$  to represent it. As a result, the local triangular feature (LTF) of the  $P_i$  is given by

$$LTF(P_i) = (TAR(P_i), TASL(P_i)). \quad (9)$$

From the definition of  $TAR$  and  $TASL$ , we can get that their feature dimensions are  $T_s$  and  $3 \times T_s$ , respectively. Therefore, we can obtain that the size of  $LTF(P_i)$  is  $1 \times D$ , where  $D = 4 \times T_s$ .

Then, we present the local contour fragment feature of a shape. For the contour segment  $C$  extracted from the shape contour  $S$ , we use three features to represent it: the curvature, tangential angle and Fourier descriptor. The contour segment  $C$  can be parameterized as  $z(t) = (x(t), y(t))$ , where the length of the curve  $C$  has been normalized to 1, and  $t$  ( $t \in [0, 1]$ ) is the parameter, which means the arc length traveled counterclockwise along the curve  $L$  starting from the starting point  $(x(0), y(0))$ .  $x(t)$  and  $y(t)$  are the horizontal and vertical coordinate of the point represented by the parameter  $t$ , respectively. The curvature  $Cur(t)$  of the curve segment  $C$  at each point is defined as

$$Cur(t) = \frac{|x'(t)y''(t) - x''(t)y'(t)|}{(x'(t)^2 + y'(t)^2)^{\frac{3}{2}}}. \quad (10)$$

We first use B-spline to fit the contour segment when calculating the curvature of the curve segment  $C$ . Then, we calculate the first derivative  $x'(t)$  and  $y'(t)$  and the second derivative  $x''(t)$  and  $y''(t)$  of the curve segment  $C$ . Finally, we calculate the curvature of each point of the curve segment  $C$  according to Eq. (6). The tangential angle  $TanAngle(t)$  of the curve segment  $C$  is defined as

$$TanAngle(t) = \arctan\left(\frac{y(t) - y(t-1)}{x(t) - x(t-1)}\right), \quad (11)$$

where  $\arctan(\cdot)$  represents the arctangent value. Finally, we give the Fourier descriptor of the curve segment  $C$ . Here, we use the normalized complex coordinates (NCC) signature method proposed by Sokic and Konjicija (2014). This method is very simple and very effective for the feature representation of open curve segment. Through subsequent experiments, we find that the Fourier descriptor has a certain complementarity with curvature and tangential angle features. As a result, it can also enhance the accuracy of shape recognition. Below, we briefly review the NCC signature method. First, we can uniformly sample the curve segment  $C$  to obtain  $N_s$  points. For each point  $P_n = (x_n, y_n)$  ( $n \in [1, N_s]$ ) of the curve segment  $C$ , we use a complex coordinate signature  $Z_n = x_n + jy_n$  to represent it. Then, we use the discrete Fourier transform on  $Z_n$ , which is given by

$$a_n = \frac{1}{N_s} \sum_{k=0}^{N_s-1} Z_k e^{-j2\pi nk/N_s} \quad (12)$$

where  $n = 0, 1, \dots, N_s$ . The Fourier coefficient  $a_n$  is used to obtain the Fourier descriptor of the contour segment  $C$ . To obtain scale invariance, we introduce a scale factor  $SC = \sum_{i=1}^{N_s-1} |a_i|$ . We only use  $Q$  Fourier coefficients to characterize the feature of contour segment  $C$ , which can be expressed as

$$FD(S) = \left\{ \frac{|a_1|}{SC}, \dots, \frac{|a_{Q/2}|}{SC}, \frac{|a_{N_s-Q/2}|}{SC}, \dots, \frac{|a_{N_s-1}|}{SC} \right\}, \quad (13)$$

where the dimension of the  $FD$  is  $1 \times Q$ . As a result, the local feature of the  $C$  is expressed as  $\{Cur(t), TanAngle(t), FD(S)\}$ , and its feature size is  $(2N_s + Q)$ . In the experiment, we sample the curve segment with  $N_s = 100$ , and set the Fourier coefficient  $Q = 32$ , so the feature size of the curve segment  $C$  is 232.

#### 4.3. Learning the multi-level contour combination features

After we extract the local triangular feature and the local contour segment feature of a shape, then we use the Fisher vector (FV) method to separately encode the local triangular feature and the local contour segment feature to obtain the high-level shape feature. The Fisher Vector method (Perronnin et al., 2010; Sánchez et al., 2013) is an effective local feature encoding method, which can effectively accumulate local features to obtain a high-level global feature representation. In the following, we briefly describe how to employ this encoding approach for 2D shape recognition.

We take the local triangular feature of the shape boundary as an example to describe the Fisher Vector coding method to construct high-level shape feature. The same is true for the local contour segment, which is equivalent to changing the local triangular feature to the local

contour segment feature. For a given boundary  $S$ , we can get the  $N$  local triangular features  $LTFS = \{f_i \in R^D, i = 1, 2, \dots, N\}$ . First, we use Fisher vector directly to encode the local triangular features and obtain its gradient vector  $G_\lambda^S$ . The gradient vector  $G_\lambda^S$  is obtained by concatenating the  $G_{\mu,j}^S$  and  $G_{\sigma,j}^S$  for  $j = 1, \dots, K$ . The  $G_\lambda^S$  is the high-level global feature of the local triangular features  $LTFS$  of shape contour  $S$  after encoding by Fisher vector, and its feature dimension is  $2KD$ .

Then, we adopt the power normalization (Perronnin et al., 2010) to suppress the burstiness problem, which uses the element-wise nonlinear operation. For each component  $f_{s_j}$  ( $j = 1, \dots, 2KD$ ) of the high-level shape feature  $FS_\lambda^S$ , the element-wise nonlinear operation is given by

$$f_{s_j} := |f_{s_j}|^\alpha \times \text{sign}(f_{s_j}), \quad (14)$$

where the value of  $\alpha$  is in the range of 0 to 1. In the experiment, the  $\alpha$  is set to 0.2. Next, we adopt the  $L_2$  normalized approach to acquire the final high-level characteristic for shape recognition. Similarly, we use the FV approach to encode the local contour segment to acquire the high-level shape feature  $FC_\lambda^S$  with a size  $2KL$ , where  $L$  is the dimension of the local contour segment feature. Finally, we combine the above two features as our multi-level contour combination feature, and use the multi-level combination feature for shape recognition. To further decrease the dimensions of features and also reduce the amount of memory usage, we use the principal component analysis (PCA) method to reduce the dimension of multi-level contour combination feature.

#### 4.4. Shape classification based on multi-level contour combination features

Through Section 4.3, we can obtain the multi-level contour combination feature for each shape. Next, we adopt the linear SVM model for shape recognition. We use the SVM approach proposed by Grammer et al. (Grammer and Singer, 2001) for classification. Let  $\{(F_i, y_i)\}_{i=1}^M$  is the training data, which contains  $M$  shapes from  $C$  classes. The  $F_i$  and  $y_i$  denote the multi-level contour combination characteristic and the class label of  $i$ th shape, respectively. Then, we train a multi-class SVM model, which is given by

$$\min_{w_1, \dots, w_C} \sum_{j=1}^M \|w_j\|^2 + \beta \sum_i \max(0, 1 + w_{l_i}^T F_i - w_{y_i}^T F_i), \quad (15)$$

where  $l_i = \arg \max_{l \in \{1, 2, \dots, C\}, l \neq y_i} w_l^T F_i$ . In Eq. (14), the left part is a regularization term, and the right part is the hinge-loss. The parameter  $\beta$  controls the relative weight of the regularization term. We adopt the LibLinear proposed by Fan et al. (Fan et al., 2008) to solve Eq. (14). In the testing stage, the class label of shape feature  $F$  is predicted by

$$\hat{y} = \arg \max_{l \in \{1, \dots, C\}} w_l^T F. \quad (16)$$

We choose to use the linear SVM model, mainly considering the high classification efficiency of the model. Meanwhile, the model can also be easily generalized to large-scale shape retrieval and classification tasks. In addition, other non-linear kernels, like the intersection kernel and Gaussian kernel function, can further improve the accuracy of shape recognition.

## 5. Experimental results

In this section, we tested the proposed method on some benchmark shape datasets and compared it to the prior state-of-the-art shape recognition. First of all, we introduced the setup of our experiment. Next, we described the recognition results of our method on three benchmark shape datasets, including the Animal (Bai et al., 2009), MPEG-7 (Latecki et al., 2000) and ETH-80 (Leibe and Schiele, 2003) datasets. Finally, we studied the effect of some parameters of our approach on the accuracy of shape recognition.

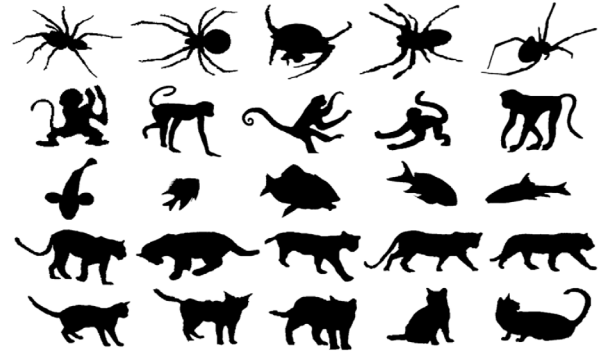


Fig. 4. Some examples of the Animal dataset. The first line presents five examples of “spider” category, the second line presents five examples of “monkey” category, the third line presents five examples of “fish” category, and the fourth and last line represent five examples of “leopard” and “cat” categories, respectively.

### 5.1. Experimental setup

We sampled  $N = 1024$  points for a given shape  $S$ , and the feature dimension of each point in the  $S$  was  $L = 4 \times T_s = 4 \times \lceil \log_2(N/2) \rceil = 4 \times 9 = 36$ . We considered each point of the  $S$  as a local characteristic of the shape. For a local contour segment extracted from a shape  $S$ , we sampled  $N_s = 100$  points. We used the curvature, tangential angle and Fourier descriptor to characterize the local contour segment. Thus, we obtained that the feature size of the local contour segment was 232. The number of clustering center  $K$  was defined as 256 when using the Fisher Vector method to encode sampling point features and local contour segment features. After the FV encoding, the dimensions of the high-level shape features were  $2KD = 2 \times 256 \times 36 = 18,432$  and  $2KL = 2 \times 256 \times 232 = 118,732$ , respectively. Then, we used the PCA approach to decrease the dimension of their characteristics to  $\hat{D} = 256$ , and we integrated the two features after the PCA dimension reduction to obtain the multi-level contour combination feature, which has a feature size of 512. Finally, we adopted a linear SVM model for shape recognition using the multi-level contour combination feature, and we set the weight of parameter  $\beta = 10$  in the experiment. We adopted the MATLAB language to implement the complete algorithm, and we employed a Intel Core i7-7500U 2.9 GHz CPU and 8 GB RAM as the experimental platform.

### 5.2. Animal dataset

We first assessed the classification performance of our approach on the Animal dataset (Bai et al., 2009), which consists of 20 classes, with 100 shapes in each class for a total of 2000 shapes. Fig. 4 presents some examples of the Animal dataset. Considering that the shapes in this dataset have complex deformations and intra-class changes, it is very difficult for the shape recognition task. Therefore, it is very suitable for evaluating the effectiveness and robustness of various shape recognition methods. We can also see from some samples in Fig. 4 that the shapes of the “leopard” class in the fourth row are very similar to the shapes of the “cat” class in the last row, which makes it much more difficult to identify these two categories. To directly compare with the other approaches, we use the same evaluation criteria as in Sun and Super (2005), Bai et al. (2009), Wang et al. (2014), Ramesh et al. (2015), Bicego and Lovato (2016), Yang et al. (2020). We randomly select 50 shapes of each class as the training set, and the rest shapes as the test set. To avoid the deviation caused by randomness, we repeated the experiment 10 times. Then we take the average value of 10 times as the classification accuracy. Table 1 lists the comparison results of our approach and the other shape recognition methods.

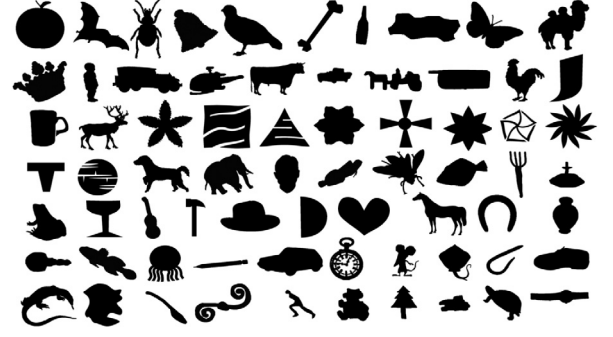
We can observe that our approach obtains the highest recognition accuracy among all the competing approaches. Compared to the deep

**Table 1**  
Classification results of different methods on Animal dataset.

| Method                                            | Classification accuracy |
|---------------------------------------------------|-------------------------|
| Class segment set (Sun and Super, 2005)           | 69.70%                  |
| IDSC (Ling, 2007)                                 | 73.60%                  |
| Bag of SIFT (Lim and Galoogahi, 2010)             | 74.90%                  |
| Contour segments (Bai et al., 2009)               | 71.70%                  |
| Skeleton paths (Bai et al., 2009)                 | 67.90%                  |
| ICS (Bai et al., 2009)                            | 78.40%                  |
| MFD (Yang and Yu, 2019)                           | 67.35%                  |
| Distinctive Parts (Li et al., 2011)               | 80.70%                  |
| BCF (Wang et al., 2014)                           | 83.40 $\pm$ 1.30%       |
| Bioinformatics (Bicego and Lovato, 2016)          | 83.70%                  |
| pSGLD (Li et al., 2016)                           | 84.80 $\pm$ 0.35%       |
| Contextual BOW (Ramesh et al., 2015)              | 86.00%                  |
| BoSCP-LP (Shen et al., 2018)                      | 89.77 $\pm$ 0.65%       |
| AlexNet + Fine-tuned (Krizhevsky et al., 2012)    | 84.10%                  |
| VGG16 + Fine-tuned (Simonyan and Zisserman, 2015) | 88.00%                  |
| ResNet50 + Fine-tuned (He et al., 2016)           | 89.50%                  |
| Informative Dropout (Shi et al., 2020)            | 78.73%                  |
| CNN + Data Augmentation (Patel et al., 2019)      | 91.65 $\pm$ 0.87%       |
| CMLR ( $\hat{D} = 2048$ ) (Yang et al., 2020)     | 89.26 $\pm$ 0.74%       |
| CMLR ( $\hat{D} = 256$ ) (Yang et al., 2020)      | 88.68 $\pm$ 0.41%       |
| <b>Our method (Point)</b>                         | 89.31 $\pm$ 0.71%       |
| <b>Our method (Contour)</b>                       | 78.02 $\pm$ 1.04%       |
| <b>Our method (Contour+Point)</b>                 | 92.70 $\pm$ 0.75%       |

learning-based approaches pSGLD (Li et al., 2016) and CNN based on data augmentation (Patel et al., 2019), the recognition result of our method is higher than them by nearly 8% and 1%, respectively. Compared with the transfer learning methods based on AlexNet (Krizhevsky et al., 2012), VGG16 (Simonyan and Zisserman, 2015) and ResNet (He et al., 2016) deep models, the classification accuracy of our method is nearly 6.6%, 4.7% and 3.2% higher than them, respectively. Compared with the shape-bias based Informative Dropout method (Shi et al., 2020), the classification accuracy of our method is nearly 14% higher than that of this method. Compared to the classic model-based methods BCF (Wang et al., 2014) and Contextual Bow (Ramesh et al., 2015), the recognition result of our approach is 9% and 6.5% higher than the two approaches, respectively. Compared to the latest model-based approaches BoSCP-LP (Shen et al., 2018) and CMLR (Yang et al., 2020), the recognition result of our method is also higher than these two methods by nearly 3% and 3.5%, respectively. In particular, we can also notice that the recognition accuracy of our method with only  $\hat{D} = 256$  dimensions of sampling point features is higher than that of CMLR method (Yang et al., 2020) using  $\hat{D} = 2048$  dimensions. At the same time, the recognition accuracy of our method using only sampling point features is higher than that of the CMLR method using only 256 feature dimensions. This shows that there is a certain feature redundancy in the description of sampling points in CMLR method, which affects the accuracy of shape recognition. Compared with the best exemplar-based approach of Bioinformatics (Bicego and Lovato, 2016), the recognition result of our approach is higher than this method by 9%. We can also notice that the classification rate of our method is 24% higher than that of the MFD approach (Yang and Yu, 2019). This shows that the low-level shape feature is not enough to effectively distinguish the shapes of the objects on the Animal dataset due to large deformation and intra-class changes. We use the 1-NN model to acquire the recognition result of the MFD method. If not specifically stated, the subsequent classification results of the MFD method are also acquired by the 1-NN model. At the same time, we can also notice that the recognition accuracy of our approach can be further improved when combining contour segment features and sampling point features, which shows that the two features are complementary. In addition, we can also see that the recognition accuracy of high-level shape feature constructed based on local sampling points is better than the shape feature methods based on contour segments.

We also give the recognition result of each category on the Animal dataset, which is presented in Table 2. It can be seen that the



**Fig. 5.** Some examples of the MPEG-7 dataset. One shape per category is given.

recognition accuracy of our method in all categories exceeds the classic ICS method (Bai et al., 2009) and BCF method (Wang et al., 2014). Compared with the CMLR method (Yang et al., 2020), the recognition accuracy of our method surpasses this method in all categories except for the horse category. This proves that the proposed method in this paper has a strong shape discrimination ability. At the same time, we can also see that our method can greatly improve the classification performance of cat, leopard and monkey categories. This shows that our method can well characterize large intra-class changes and nonlinear deformations of an object's shape. Hence, we believe that our approach is more suitable for the task of shape recognition than the other approaches.

### 5.3. MPEG-7 dataset

Then we test the recognition performance of our approach on the MPEG-7 dataset (Latecki et al., 2000), which is a famous dataset for 2D shape analysis in pattern recognition. This dataset contains a total of 1400 shapes, separated into 70 categories, each category has 20 shapes. Fig. 5 presents some examples on the MPEG-7 dataset. Some shapes in this dataset have internal structures and complex deformations. Therefore, this dataset is often used to evaluate the performance of various shape descriptors.

Some shape recognition approaches have been tested on the MPEG-7 dataset (Sun and Super, 2005; Wang et al., 2014; Patel et al., 2019). To directly compare the performance with these methods, we use the same evaluation metric, that is, we randomly choose half of the images of each category for training, and the remaining images are used for testing. We repeated the experiment 10 times, and used the average value of the 10 experiments as the classification accuracy. Table 3 reports the classification results of our approach and the other shape recognition approaches.

We can observe that our approach obtains the best classification result among all the approaches to be compared. Compared to the deep learning-based method of CNN based on data augmentation (Patel et al., 2019), the classification rate of our approach is 2% higher than this method. Compared with the transfer learning methods based on AlexNet (Krizhevsky et al., 2012), VGG16 (Simonyan and Zisserman, 2015) and ResNet (He et al., 2016) deep models, the classification accuracy of our method is nearly 9%, 5.5% and 11% higher than them, respectively. Compared with the shape-bias based Informative Dropout method (Shi et al., 2020), the classification accuracy of our method is nearly 2.5% higher than that of this method. Compared with the example-based methods of Bioinformatics (Bicego and Lovato, 2016) and multiscale Fourier descriptor (Yang and Yu, 2019), the classification rate of our approach is higher than them by nearly 3.1% and 3%, respectively. Compared with the model-based BoSCP-LP (Shen et al., 2018) method, the classification accuracy of our method is still 0.8% higher than this method. Even compared with the recently proposed advanced model-based CMLR (Yang et al., 2020) method, the

**Table 2**

The classification accuracy of different methods for each class on the Animal dataset.

| Method                   | Bird         | Butterfly    | Cat          | Cow          | Crocodile    | Deer         | Dog          | Dolphin      | Duck         | Elephant     |
|--------------------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|--------------|
| CS (Bai et al., 2009)    | 0.760        | 0.890        | 0.390        | 0.700        | 0.540        | 0.690        | 0.690        | 0.870        | 0.830        | 0.950        |
| ICS (Bai et al., 2009)   | 0.760        | 0.930        | 0.480        | 0.800        | 0.660        | 0.790        | 0.750        | 0.890        | 0.890        | 0.970        |
| BCF (Wang et al., 2014)  | 0.876        | 0.922        | 0.738        | 0.774        | 0.768        | 0.904        | 0.826        | 0.890        | 0.870        | 0.952        |
| CMLR (Yang et al., 2020) | 0.912        | 0.964        | 0.820        | 0.820        | 0.858        | 0.922        | 0.820        | 0.950        | 0.936        | 0.956        |
| <b>Our method</b>        | <b>0.960</b> | <b>1.000</b> | <b>0.830</b> | <b>0.900</b> | <b>0.960</b> | <b>0.960</b> | <b>0.900</b> | <b>0.960</b> | <b>0.980</b> | <b>0.980</b> |
| Method                   | Fish         | Flyingbird   | Hen          | Horse        | Leopard      | Monkey       | Rabbit       | Rat          | Spider       | Tortoise     |
| CS (Bai et al., 2009)    | 0.700        | 0.570        | 0.890        | 0.960        | 0.560        | 0.210        | 0.810        | 0.520        | 0.980        | 0.810        |
| ICS (Bai et al., 2009)   | 0.740        | 0.650        | 0.940        | 0.970        | 0.650        | 0.330        | 0.870        | 0.840        | 1.000        | 0.900        |
| BCF (Wang et al., 2014)  | 0.798        | 0.720        | 0.942        | 0.954        | 0.664        | 0.584        | 0.858        | 0.706        | 0.992        | 0.936        |
| CMLR (Yang et al., 2020) | 0.812        | 0.848        | 0.976        | <b>0.992</b> | 0.802        | 0.750        | 0.970        | 0.796        | 0.996        | 0.952        |
| <b>Our method</b>        | <b>0.820</b> | <b>0.920</b> | <b>0.960</b> | 0.980        | <b>0.820</b> | <b>0.770</b> | <b>0.980</b> | <b>0.860</b> | <b>1.000</b> | <b>1.000</b> |

**Table 3**

Classification results of different approaches on MPEG-7 dataset.

| Method                                            | Classification accuracy             |
|---------------------------------------------------|-------------------------------------|
| Class segment set (Sun and Super, 2005)           | 90.9%                               |
| IDSC+DP (Ling, 2007)                              | 88.43%                              |
| Contour segments (Bai et al., 2009)               | 91.10%                              |
| Skeleton paths (Bai et al., 2009)                 | 86.70%                              |
| ICS (Bai et al., 2009)                            | 96.60%                              |
| BCF (Wang et al., 2014)                           | 97.16% $\pm$ 0.79%                  |
| Bioinformatics (Bicego and Lovato, 2016)          | 96.10%                              |
| MFD (Yang and Yu, 2019)                           | 96.27%                              |
| BoSCP-LP (Shen et al., 2018)                      | 98.41 $\pm$ 0.44%                   |
| AlexNet + Fine-tuned (Krizhevsky et al., 2012)    | 90.29%                              |
| VGG16 + Fine-tuned (Simonyan and Zisserman, 2015) | 93.71%                              |
| ResNet50 + Fine-tuned (He et al., 2016)           | 88.00%                              |
| Informative Dropout (Shi et al., 2020)            | 96.90%                              |
| CNN+Data Augmentation (Patel et al., 2019)        | 97.24 $\pm$ 0.51%                   |
| CMLR (Yang et al., 2020)                          | 98.77 $\pm$ 0.65%                   |
| <b>Our method (Point)</b>                         | <b>98.83 <math>\pm</math> 0.45%</b> |
| <b>Our method (Contour)</b>                       | <b>95.31 <math>\pm</math> 0.80%</b> |
| <b>Our method (Contour+Point)</b>                 | <b>99.26 <math>\pm</math> 0.43%</b> |

classification accuracy of our method is also better than this method. This experiment further proves that our approach has a strong ability of shape discrimination. At the same time, we also notice that the accuracy of shape classification can be improved by effectively combining the features of contour sampling points and contour segments. This shows that the contour point feature and the contour segment feature have strong complementarity.

#### 5.4. ETH-80 dataset

Finally, we test the recognition performance of our approach on the ETH-80 dataset (Leibe and Schiele, 2003). This dataset contains a total of 80 high-resolution objects, divided into eight categories. Each object has 41 color images with different angles, so there are 3280 images in total. We employ the segmentation masks of all images supplied by this dataset to test the performance of various shape recognition methods. Fig. 6 shows some samples of this dataset.

This dataset used the leave-one-object-out performance metric to test the accuracy of shape recognition, which was proposed in Leibe and Schiele (2003). This performance metric indicates that all shapes from 79 objects are employed as the training set, and the shapes of the rest object are applied to the testing set. We compare the recognition result of our approach with that of the other shape recognition approaches, and Table 4 presents the comparison results of our approach and other shape recognition methods.

We can notice that our approach obtains the best classification result among all the compared methods. Compared to the deep learning approach of CNN based on hierarchical characteristic (Ke and Li, 2014), the classification result of our approach is 2.5% higher than this method. Compared with the transfer learning methods based on AlexNet (Krizhevsky et al., 2012), VGG16 (Simonyan and Zisserman, 2015) and ResNet (He et al., 2016) deep models, the classification

**Fig. 6.** Some examples of the ETH-80 dataset. Each class is presented in each row.**Table 4**

Classification rates of different approaches on ETH-80 dataset.

| Method                                            | Classification accuracy             |
|---------------------------------------------------|-------------------------------------|
| Color histogram (Leibe and Schiele, 2003)         | 64.86%                              |
| PCA gray (Leibe and Schiele, 2003)                | 82.99%                              |
| PCA masks (Leibe and Schiele, 2003)               | 83.41%                              |
| MTCD (Yang et al., 2016)                          | 82.29%                              |
| SC+DP (Leibe and Schiele, 2003)                   | 86.40%                              |
| MFD (Yang and Yu, 2019)                           | 86.91%                              |
| IDSC+DP (Ling, 2007)                              | 88.11%                              |
| IDSC+Morphological strategy (Hu et al., 2012)     | 88.04%                              |
| Height function (Wang et al., 2012)               | 88.72%                              |
| Robust symbolic (Daliri and Torre, 2008)          | 90.28%                              |
| Kernel-edit distance (Daliri and Torre, 2010)     | 91.33%                              |
| BCF (Wang et al., 2014)                           | 91.49%                              |
| Bioinformatics (Bicego and Lovato, 2016)          | 91.50%                              |
| Manifold MKL SVM (Jayasumana et al., 2013)        | 93.75%                              |
| AlexNet + Fine-tuned (Krizhevsky et al., 2012)    | 88.16%                              |
| VGG16 + Fine-tuned (Simonyan and Zisserman, 2015) | 90.14%                              |
| ResNet50 + Fine-tuned (He et al., 2016)           | 85.01%                              |
| Informative Dropout (Shi et al., 2020)            | 98.21%                              |
| Hierarchical feature using CNN (Ke and Li, 2014)  | 95.80%                              |
| CMLR (Yang et al., 2020)                          | 97.93%                              |
| <b>Our method (Point)</b>                         | <b>98.14 <math>\pm</math> 1.35%</b> |
| <b>Our method (Contour)</b>                       | <b>95.24 <math>\pm</math> 2.12%</b> |
| <b>Our method (Contour+Point)</b>                 | <b>98.32 <math>\pm</math> 1.43%</b> |

accuracy of our method is nearly 10.2%, 8.2% and 13% higher than them, respectively. Compared with the shape-bias based Informative Dropout method (Shi et al., 2020), the classification accuracy of our method is slightly higher than that of this method. Compared with the classic model-based BCF method (Wang et al., 2014), the classification result of our approach is 6.8% higher than this method. Meanwhile, compared with the best model-based CMLR approach (Yang et al., 2020), the classification result of our approach is 0.4% higher than this

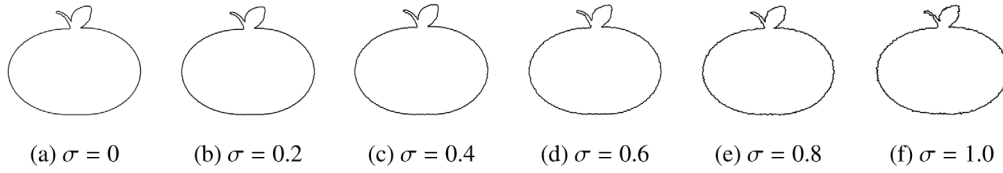


Fig. 7. An example of shape boundaries under different noise ratios  $\sigma$ .

method. Compared to the best example-based approach of Bioinformatics (Bicego and Lovato, 2016), the classification result of our approach is 6.8% higher than this method. We can also see that the classification result of our approach is also higher than the other approaches when our method only uses sampling point features. This proves the effectiveness and robustness of our approach. Meanwhile, we can also see that the classification result obtained using the point feature is higher than that based on the contour feature. The possible reason is that a great deal of redundant contour fragments will be generated when dividing the complete contour by the DCE method, which will affect the performance of shape recognition. Similarly, the accuracy of shape recognition will be further improved when the sampling point feature and contour segment feature are combined. This shows that these two characteristics are complementary.

##### 5.5. Robustness to noise

We can notice that the shape contours are very smooth in the above experiments. However, in practice, a very complete or smooth contour cannot be obtained due to segmentation errors or noise. To assess the recognition performance of our method under the noise addition, we add Gaussian noise to the shape boundary to further analyze the robustness of our method. We employ the whole MPEG-7 dataset (Latecki et al., 2000) as the initial shape boundary. These noises are added by changing the  $x$ - and  $y$ -coordinates of the boundary. The noise value comes from a Gaussian function with zero mean and  $\sigma$  variance. We present an instance of shape boundaries with different Gaussian noise, which is presented in Fig. 7. It can be seen that the shape contour becomes rougher as the value of the Gaussian noise  $\sigma$  increases.

We use the performance evaluation standard of the MPEG-7 dataset, which is explained in 4.3 section. Fig. 8 presents the recognition results of our approach under different noise levels. We can observe that the recognition result of our method is almost unchanged for the noise ratio  $\sigma = 0.2$ . Meanwhile, the recognition result of our approach also remains stable with the noise from  $\sigma = 0.4$  to  $\sigma = 0.6$ . In addition, our method still obtains a high recognition accuracy when the noise ratio  $\sigma$  changes from 0.8 to 1. The recognition result of our approach is only reduced by 0.5% while  $\sigma$  increases from 0 to 1. As a result, our method is very robust to noise.

##### 5.6. Parameter study

In this subsection, we further analyze the effect of the parameters in our approach on the accuracy of shape recognition. We use the Animal dataset for the parameter analysis. Our method mainly has six parameters: the number of contour sampling points  $N$ , the number of sampling points for contour fragment  $N_c$ , the Fourier coefficients  $Q$  of the NCC feature, the threshold parameter  $\tau$  of the DCE method, the number of clusters  $K$  and the dimension  $\hat{D}$  of the PCA. Among them, the parameters of  $N$ ,  $N_s$ ,  $Q$  and  $\tau$  affect the performance of low-level shape representation. As a result, we not only explore the final shape recognition performance of the multi-level contour combined feature, but also give the recognition accuracy of their respective features on the Animal dataset. For the number of clustering center  $K$  and the dimension  $\hat{D}$  of the PCA, we only give the final shape recognition performance of the multi-level contour combination feature on the

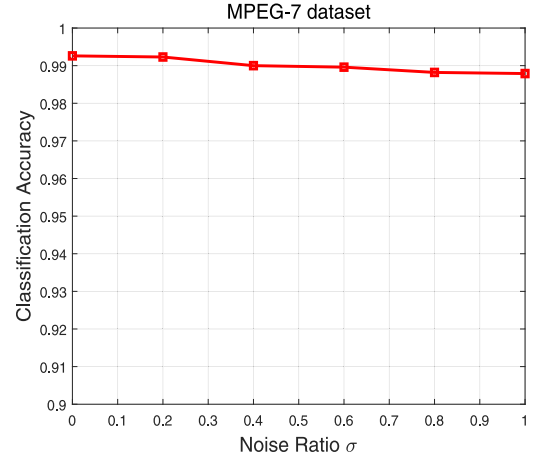


Fig. 8. Recognition results of the MPEG-7 dataset are shown with the noise ratio  $\sigma$  varying from 0 to 1.

Animal dataset. At the same time, other parameters remain unchanged when we analyze a certain parameter.

First, we analyze the parameters of contour point feature and contour segment feature. For the feature representation of contour points, the parameter of our method is contour sampling point  $N$ , and we take five values for  $N$ : 64, 128, 256, 512 and 1024. Fig. 9 shows the recognition performance of our approach on the Animal dataset for five different boundary points  $N$ . We can see that the recognition result of our approach increases as  $N$  increases. The best recognition result is achieved when  $N = 1024$ . Meanwhile, the classification accuracy of our method decreases when  $N$  is greater than 1024, which means that approximately 1000 local features can effectively describe the shape information of an object. If excessive local features are used, it will affect the accuracy of shape recognition due to the feature redundancy.

For the feature representation of the contour segment, there are three parameters: the sampling point  $N_s$  of the contour segment, the Fourier coefficients  $Q$  and the threshold parameter  $\tau$  of the DCE method. Among them, we take four values for  $N_s$ : 50, 100, 150, 200, and take four values for Fourier coefficients  $Q$ : 8, 16, 32, 64. For the parameter  $\tau$  of the DCE method, we take five values: 0.1, 0.2, 0.3, 0.4 and 0.5. Fig. 10(a)–(c) shows the recognition accuracy of our method under different  $N_c$ ,  $Q$  and  $\tau$  values. Among them, the red curve indicates that our method uses the contour combination feature for shape recognition, that is, combining contour segment feature and sampling point feature, and the green curve indicates that our method only uses contour segment feature for shape recognition.

First, we can see from Fig. 10(a) that our approach obtains the best classification result when  $N_s = 100$ . Meanwhile, we can observe that the classification result of our approach first increases and then decreases with the increase of  $N_s$ . However, we can notice that the overall classification result of our method remains relatively stable with the change of  $N_s$ . Then, we can see from Fig. 10(b) that the best classification result of our method is obtained when  $Q = 32$ . The classification accuracy of our method decreases first, then increases and then decreases as  $Q$  increases. Nevertheless, the overall classification

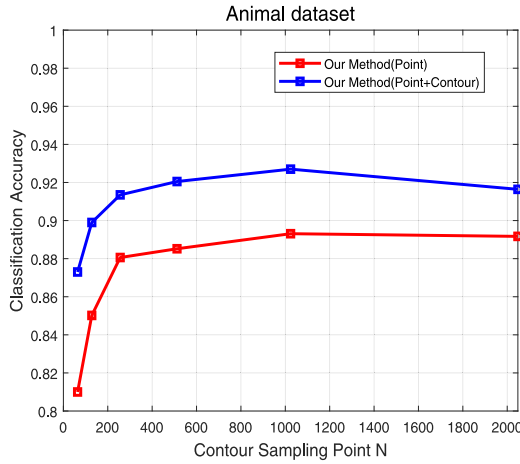


Fig. 9. Recognition results of the Animal dataset are shown at five different sampling points  $N$ .

accuracy of our approach remains unchanged. That is, the Fourier coefficient  $Q$  has a relatively low impact on the overall performance of shape recognition. Finally, we can see from Fig. 10(c) that the classification accuracy of our method based on the contour combination feature increases first and then decreases as the parameter  $\tau$  increases, but the classification accuracy of our method based on contour segment feature decreases with the increase of parameter  $\tau$ . However, the classification accuracy of our method based on the contour combination feature remains stable with the change of parameter  $\tau$ , which shows that the parameter  $\tau$  only affect the accuracy of our method based on the contour segment feature. In addition, we can see from Fig. 10(a)–(c) that the overall classification accuracy of our approach remains stable with the change of any parameters in the contour segment feature, which further demonstrates the robustness of our method.

Next, we analyze the influence of the number of clustering center  $K$  in FV encoding on the performance of shape recognition. We take five values for clustering center  $K$ : 64, 128, 256, 512 and 1024. Fig. 11 shows the recognition result of our approach on the Animal dataset under different number of clustering center  $K$ . We can observe that our method obtains the best classification accuracy when  $K = 256$ . Meanwhile, we can notice that the recognition result of our approach first increases and then decreases as the  $K$  value increases. Therefore, we set  $K = 256$  in the experiment.

Finally, we analyze the influence of the PCA dimension reduction method on the performance of our method. Since the dimension after FV encoding is particularly high, while the PCA is an effective dimension reduction method to reduce the memory usage and improve efficiency. We reduce the dimensions of the contour segment feature and the sampling point feature after the FV encoding, respectively. Meanwhile, we reduce the dimensions to the same size, and then combine them as the multi-level contour combination feature. We take five values for the size  $\hat{D}$  of the PCA method: 64, 128, 256, 512 and 1024. Fig. 12 shows the recognition result of our approach under different dimension reduction sizes  $\hat{D}$ . We can observe that the recognition result of our approach first increases and then decreases as the feature dimension  $\hat{D}$  increases. Meanwhile, our method achieves the best classification accuracy when  $\hat{D} = 256$ . Therefore, our method obtains a lower feature dimension, which can improve the efficiency of recognition and reduce memory usage. Therefore, the multi-level contour combination feature we constructed has a lower feature dimension, which can improve the efficiency of recognition and reduce memory usage. In all experiments, we reduced the feature dimension of our method to 256.

Table 5

Classification results of local contour sampling points based on different triangular feature combinations on the Animal dataset.

| Method              | Classification accuracy             |
|---------------------|-------------------------------------|
| Point(TAR)          | 79.22 $\pm$ 1.07%                   |
| Point(TASL)         | 84.85 $\pm$ 0.70%                   |
| Point(TCD)          | 73.23 $\pm$ 1.48%                   |
| Point(TAR+TCD)      | 86.00 $\pm$ 1.28%                   |
| Point(TASL+TCD)     | 86.27 $\pm$ 0.63%                   |
| Point(TAR+TASL)     | <b>89.31 <math>\pm</math> 0.71%</b> |
| Point(TAR+TASL+TCD) | 88.68 $\pm$ 0.41%                   |

Table 6

Classification results of different feature combinations of contour segments on the Animal dataset.

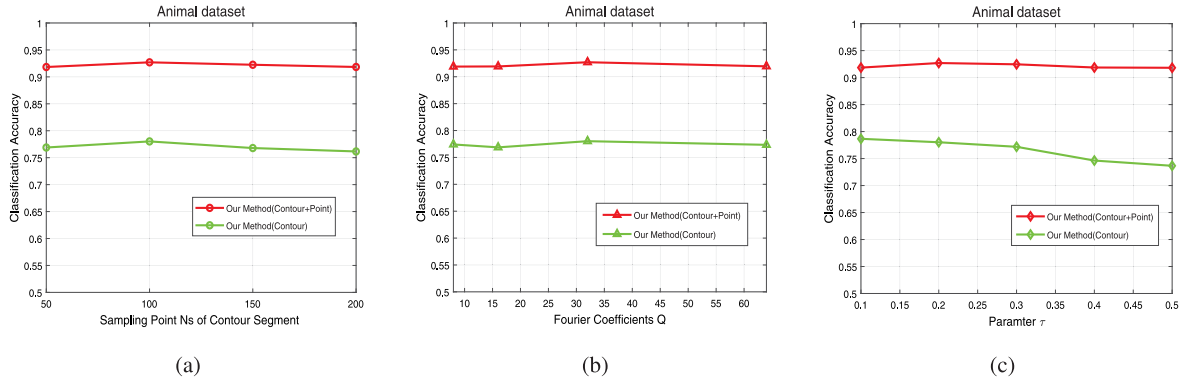
| Method                           | Classification accuracy             |
|----------------------------------|-------------------------------------|
| Contour( $Cur$ )                 | 70.32 $\pm$ 1.15%                   |
| Contour( $TanAngle$ )            | 73.53 $\pm$ 1.43%                   |
| Contour( $FD$ )                  | 77.13 $\pm$ 0.73%                   |
| Contour( $Cur + TanAngle$ )      | 76.62 $\pm$ 0.95%                   |
| Contour( $Cur + FD$ )            | 73.98 $\pm$ 1.01%                   |
| Contour( $TanAngle + FD$ )       | 75.31 $\pm$ 1.11%                   |
| Contour( $Cur + TanAngle + FD$ ) | <b>78.02 <math>\pm</math> 1.04%</b> |

### 5.7. Ablation study

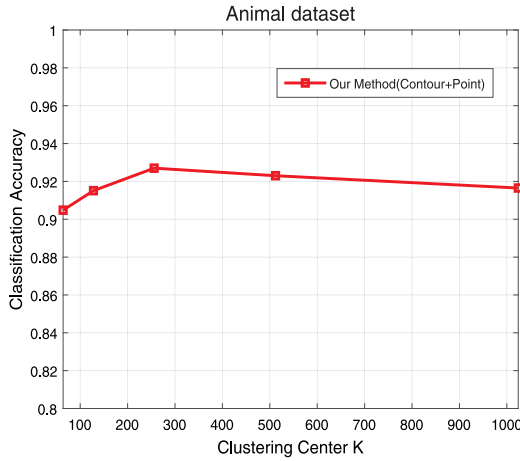
In this section, we conducted ablation experiments to study the influence of different feature combinations on the experimental results. We used the Animal dataset for the ablation study, and used the classification accuracy as the performance metric. We randomly selected half of the shape data of each class in the Animal dataset as the training set and the remaining shape data as the test set. First, we study the influence of local contour sampling points based on different triangular feature combinations on the experimental results. Table 5 shows the experimental results of different triangular feature combinations on the Animal dataset. In addition to exploring the two triangular features of TAR and TASL, we also analyzed the triangular feature of TCD. We can see that the best recognition accuracy can be achieved when two triangular features of TAR and TASL are used as local features of contour sampling points. At the same time, we notice that the classification accuracy decreases by 0.6% when the TCD feature is further introduced. The possible reason is that the combination of TCD feature with TAR and TASL features generates redundancy, which affects the recognition performance. Therefore, we only use TAR and TASL triangular features to represent contour sampling points.

Then, we further analyze the influence of different feature combinations of local contour segments on the experimental results. Among the contour segment features used in our method, we used the curvature feature ( $Cur$ ), tangent angle ( $TanAngle$ ), and Fourier descriptor ( $FD$ ) of the contour segment. Table 6 shows the classification results of different feature combinations of local contour segments on the Animal dataset. We can see that the Fourier descriptor achieves the highest recognition accuracy among the three contour segment features alone. At the same time, we also notice that the best recognition accuracy is achieved when all three contour segment combinations are combined, which indicates the strong complementarity of these three features. Therefore, we use the curvature feature, tangent angle, and Fourier descriptor as feature representations of the contour segment.

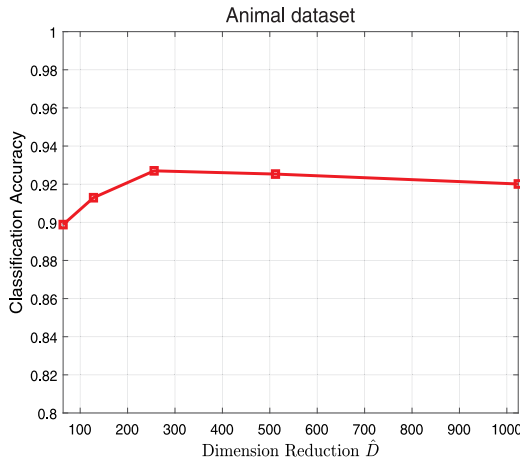
Finally, we study the influence of the combination of different sampling point features and contour segment features on the experimental results. Table 7 shows the experimental results on different combinations of sampling point features and contour segment features on the Animal dataset. It can be seen that the best classification accuracy is achieved when the sampling point-based TAR and TASL features are combined with the curvature features  $Cur$ , tangent angle  $TanAngle$  and Fourier descriptor  $FD$  of the contour segments. At the same time, we also notice that the classification accuracy increases with the addition



**Fig. 10.** Parameters analysis of contour segment feature: (a) classification result of the Animal dataset for four different sampling point  $N_s$  of the contour segment; (b) classification result of the Animal dataset at four different Fourier coefficients  $Q$ ; (c) classification result of the Animal dataset under five different parameter  $\tau$ . (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)



**Fig. 11.** Classification accuracies of the Animal dataset are shown under different number of clusters  $K$ .



**Fig. 12.** Classification accuracies of the Animal dataset are shown under different feature dimension  $\hat{D}$  of the PCA method.

of contour segment features when the TAR and TASL features of sampling points are used. In addition, the classification accuracy is also significantly improved with the addition of triangular features based on sampling points when all contour segment features are used. This shows that the performance of shape recognition is significantly improved by effectively fusing the features of sampling points and contour segments, and these five features are highly complementary. The possible reason

**Table 7**

Classification results of different combinations of sampling point features and contour segment features on the Animal dataset.

| Method                                           | Classification accuracy              |
|--------------------------------------------------|--------------------------------------|
| Point(TAR+TASL)+Contour( $Cur$ )                 | $90.53 \pm 0.54\%$                   |
| Point(TAR+TASL)+Contour( $TanAngle$ )            | $90.80 \pm 1.21\%$                   |
| Point(TAR+TASL)+Contour( $FD$ )                  | $90.75 \pm 0.96\%$                   |
| Point(TAR+TASL)+Contour( $Cur + TanAngle$ )      | $90.86 \pm 0.56\%$                   |
| Point(TAR+TASL)+Contour( $Cur + FD$ )            | $91.15 \pm 0.86\%$                   |
| Point(TAR+TASL)+Contour( $TanAngle + FD$ )       | $91.43 \pm 0.56\%$                   |
| Point(TAR)+Contour( $Cur + TanAngle + FD$ )      | $86.88 \pm 1.14\%$                   |
| Point(TASL)+Contour( $Cur + TanAngle + FD$ )     | $89.23 \pm 0.75\%$                   |
| Point(TAR+TASL)+Contour( $Cur + TanAngle + FD$ ) | <b><math>92.70 \pm 0.75\%</math></b> |

is that the multi-level contour combination features can characterize the complex variation of shapes from coarse to fine. That is, contour sampling points are used to characterize the subtle changes of a shape, and contour segments are used to characterize the global changes of a shape. Therefore, the proposed multi-level contour combination features consider the two levels of sampling points and contour segments and achieve state-of-the-art classification performance by effectively fusing them.

### 5.8. Object classification on real-world images

In all the above experiments, we assume that the contour of an object can be completely extracted from the natural image. However, it is an extremely difficult task to extract the complete contour of an object from the natural image. Because objects are often affected by factors such as noise, occlusion, and cluttered background in real scenes, resulting in partial missing of the extracted object contours. Therefore, one might wonder how the proposed method detects and recognizes objects in real-world images. In order to further prove that our method still has a strong ability to recognize objects in natural scenes. We conducted experiments on natural scene images. We combined the Weizmann Horse (Borenstein et al., 2004) and ETHZ Cow (Leibe and Schiele, 2003) datasets. Among them, the Weizmann Horse dataset has 328 horse images and the ETHZ cow dataset contains 111 cow images, with a total of 439 natural scene images. We used an unsupervised game-theoretic saliency detection method (Zeng et al., 2018) to extract the foreground region for each image. Fig. 13 shows the saliency detection results of some images. Among them, the first two columns in Fig. 13 show several typical detection results, while the other columns show several detection results with segmentation errors.

To evaluate the performance of various shape recognition methods, we randomly select 60 images for each category to train the classification model, and the remaining images are used for testing.

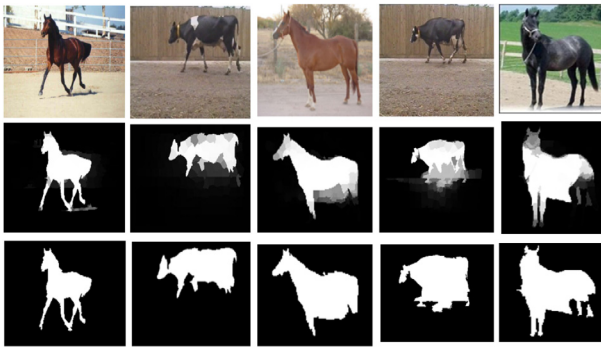


Fig. 13. The results of foreground region extraction from some images on the Weizmann Horse and ETHZ Cow datasets. The first row shows some natural scene images, the second row shows the detection results of the unsupervised game theory method, and the third row shows the binarization result of the salient maps. The first two columns represent several typical results, and the other columns indicate some unsatisfactory results of foreground extraction.

Table 8

Classification results of the Weizmann Horse and ETHZ Cow datasets.

| Method                    | Classification accuracy |
|---------------------------|-------------------------|
| MFD (Yang and Yu, 2019)   | 82.48 $\pm$ 2.05%       |
| BCF (Wang et al., 2014)   | 83.23 $\pm$ 1.50%       |
| CMLR (Yang et al., 2020)  | 93.32 $\pm$ 1.58%       |
| Our method(Point)         | 98.53 $\pm$ 0.43%       |
| Our method(Contour)       | 85.65 $\pm$ 1.83%       |
| Our method(Contour+Point) | 99.25 $\pm$ 0.32 %      |

We performed 10 experiments and used their average value as the final classification accuracy. We compare the proposed method with the BCF method (Wang et al., 2014), the multiscale Fourier descriptor method (Yang and Yu, 2019) and the CMLR method (Yang et al., 2020). The BCF and MFD methods are representative methods of mid-level shape feature and low-level shape descriptor, and the CMLR method is a recently proposed method of mid-level shape feature. Table 8 lists the comparison results of these methods. It can be seen that our method obtains the highest classification accuracy, which is 16% higher than the BCF method (Wang et al., 2014) and nearly 17% higher than the MFD method (Yang and Yu, 2019). At the same time, the classification accuracy of our method is also higher than the CMLR method by nearly 6%. In addition, we can also notice that the classification accuracy of our method using only local sampling point features is still higher than that of the BCF method, the MFD method and the CMLR method. Especially, it can be seen from the comparison results of the proposed method and the CMLR method that the CMLR method has a certain feature redundancy, which reduces the performance of shape recognition. When the contour segment feature is combined with the local point feature, the classification accuracy can be further improved. This result shows that these two characteristics are highly complementary. Therefore, this experiment proves that our method can be applied to the classification of objects in real-world scenes, and it is also robust to foreground segmentation errors and local boundary noise.

## 6. Conclusions

We have presented a new multi-level contour combination feature for shape recognition. This combination feature can tackle the intra-class changes and complex deformation of object shapes very well, and improve the performance of shape recognition. Our contribution is three-fold in comparison to previous work. First, we divide the shape contour into two levels: the contour segment and the sampling point, and introduce an effective local feature representation approach to characterize them. Second, we propose an effective shape recognition framework based on Fisher Vector encoding method that combines

the contour segment and sampling point features. Finally, the proposed multi-level contour combination feature has achieved the best recognition performance on three benchmark shape datasets, which outperforms the prior state-of-the-art shape recognition methods. In addition, our method has also achieved high classification accuracy in the object classification task of real-world images, which further prove the effectiveness and robustness of our method. Future studies will consider how to apply this approach for 3D shape recognition.

## CRedit authorship contribution statement

**Chengzhuan Yang:** Conceptualization, Methodology, Writing – original draft. **Lincong Fang:** Investigation, Software, Data curation. **Benjie Fei:** English polish, Error-checking. **Qian Yu:** Investigation, Software, Data curation. **Hui Wei:** Writing – review & editing, Supervision.

## Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Data availability

Data will be made available on request.

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